

Lab 8 (Bonus): Code Coverage and Test Quality

Objective

Learn how to measure code coverage, identify gaps in testing, and improve test quality.

Prerequisites

- Completion of **Lab 7** or familiarity with advanced mocking techniques.
- Visual Studio Enterprise or an external code coverage tool like Coverlet installed.

Instructions

Step 1: Enable Code Coverage in Visual Studio

1. Open Visual Studio.
2. Ensure the **Test** menu has the option **Analyze Code Coverage for All Tests**:
 - If not using Visual Studio Enterprise, install **Coverlet** via NuGet in the `NunitLab.UnitTests` project.

Step 2: Analyze Code Coverage for `BankAccount`

1. Run all tests in the `NunitLab.UnitTests` project.
2. Analyze code coverage by selecting **Test > Analyze Code Coverage**.
3. Review the report and identify any uncovered methods in the `BankAccount` class.

Step 3: Add Missing Tests

1. Identify gaps in the `BankAccount` tests (e.g., handling negative or zero amounts in `Deposit` and `Withdraw` methods).
 2. Add the following tests to `BankAccountTests.cs`:
-

```
[Test]
public void Deposit_WhenAmountIsNegative_ShouldThrowArgumentException()
{
    Assert.Throws<ArgumentException>(() => _account.Deposit(-50));
}

[Test]
public void
Withdraw_WhenAmountIsGreaterThanBalance_ShouldThrowInvalidOperationException()
{
    Assert.Throws<InvalidOperationException>(() => _account.Withdraw(200));
}
```

3. Re-run tests and verify the code coverage report shows improved coverage.

Step 4: Evaluate Test Quality

1. Use the following checklist to assess test quality:
 - Do tests cover all paths (happy and edge cases)?
 - Are tests clearly named to describe their intent?
 - Do tests validate all expected outcomes (e.g., results, exceptions)?
2. Refactor any poorly written tests for readability or maintainability.

Step 5: Optimize Test Coverage

1. Add `[ExcludeFromCodeCoverage]` attribute to methods that do not need testing (e.g., logging or trivial getters/setters):

```
[System.Diagnostics.CodeAnalysis.ExcludeFromCodeCoverage]
public void Log(string message)
{
    Console.WriteLine(message);
}
```

2. Re-run the coverage analysis to focus on meaningful improvements.

Outcome

Students will:

- Understand how to measure and improve code coverage.
- Identify gaps in test suites and write additional tests to achieve high-quality coverage.